

Smart Lender: Project System Architecture & Schema Design

1. Technical Topography & Operational Workflow

The system utilizes a lightweight, decoupled three-tier architecture structured explicitly to optimize machine learning ingestion tasks. Client requests originate from a presentation layer formatted using standard fluid rows, passing parameters as POST requests. The middleware container environment intercepts payloads via an asynchronous routing framework, matches features to data pre-processing indices, forwards arrays to the specified core predictor engine, and pipes final analytics blocks directly to the tracking dashboard visualization views.

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[Web Browser Form Input UI Component] | ▼ (POST Ingestion Form Request Package) [Flask Application Runtime Controller Node (app.py)] | ┌──────────┴───┐ ▼ ▼ [Pre-processing Scaler] [Multi-Model Evaluation Array Matrix (Epic 4 Context)] |─▶ Decision Tree Engine (Validation Acc: 82%) |─▶ Random Forest Layer (Validation Acc: 87%) |─▶ KNN Group Clustered (Validation Acc: 79%) |─▶ XGBoost Classifier (Validation Acc: 89% Peak) | ▼ [Relational Data Record Sink Array (PREDICTION_RESULT Layout Match)] | ▼ (Dynamic Jinja2 View Update Payload) [Interactive Risk Result Statistics Dashboard Presentation Template]
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2. Entity Relationship Diagram Model Structure

The data persistence layout maps application transaction records seamlessly onto standard relational database boundaries using the following structural components:

- **USER:** Identity access management entity tracking system operations (user_id [PK], name, email, role, created_at).
- **APPLICANT_PROFILE:** Stores customer socio-demographic indicators (applicant_id [PK], gender, married, education, self_employed, dependents, property_area).
- **CREDIT_HISTORY:** Collects institutional financial risk and score benchmarks (credit_id [PK], applicant_id [FK], credit_score, credit_history_status).
- **LOAN_APPLICATION:** Tracks requested parameters (loan_id [PK], applicant_id [FK], income, coapplicant_income, loan_amount, loan_term).
- **PREDICTION_RESULT:** The unified analytical destination sink capturing live execution logs (prediction_id [PK], loan_id [FK], model_id [FK], prediction_status, probability_score, prediction_time).